

Derbyshire Shared Care Pathology Guidelines

Hypercalcaemia

Purpose of Guideline

The management of patients with newly diagnosed hypercalcaemia

Definition

Adjusted (corrected) calcium > 2.60 mmol/L

Importance of Hypercalcaemia

Hypercalcaemia can be a medical emergency (see below). Even when calcium is not raised to acutely dangerous levels it is important to define the cause because hypercalcaemia can be a presenting feature of serious disease including malignancy.

When is hypercalcaemia considered a medical emergency?

- | | | |
|-----------|--------------------|---------------------------------|
| • Calcium | 2.60 – 3.00 mmol/L | Not usually a medical emergency |
| • Calcium | 3.00 – 3.50 mmol/L | Possible medical emergency |
| • Calcium | >3.50 mmol/L | Usually a medical emergency |

These limits are for guidance only; the severity of the patient's symptoms, and other factors, e.g. renal function, will also determine whether the patient requires emergency admission.



The laboratory usually telephones calcium results ≥ 3.50 mmol/L to GP practices when open or to Derbyshire Health United when the GP practice is closed.

When is hypercalcaemia suspected?

Mild hypercalcaemia is relatively common, and many patients may be asymptomatic. The severity and number of symptoms will be dependent on the degree of hypercalcaemia. Patients who have developed hypercalcaemia acutely are also much more likely to have symptoms.

Symptoms include;

- Polyuria and polydipsia (most specific symptom)
- Nausea and vomiting
- Constipation
- Weakness and tiredness
- Depression Drowsiness and confusion
- Bone Pain
- Renal Stones

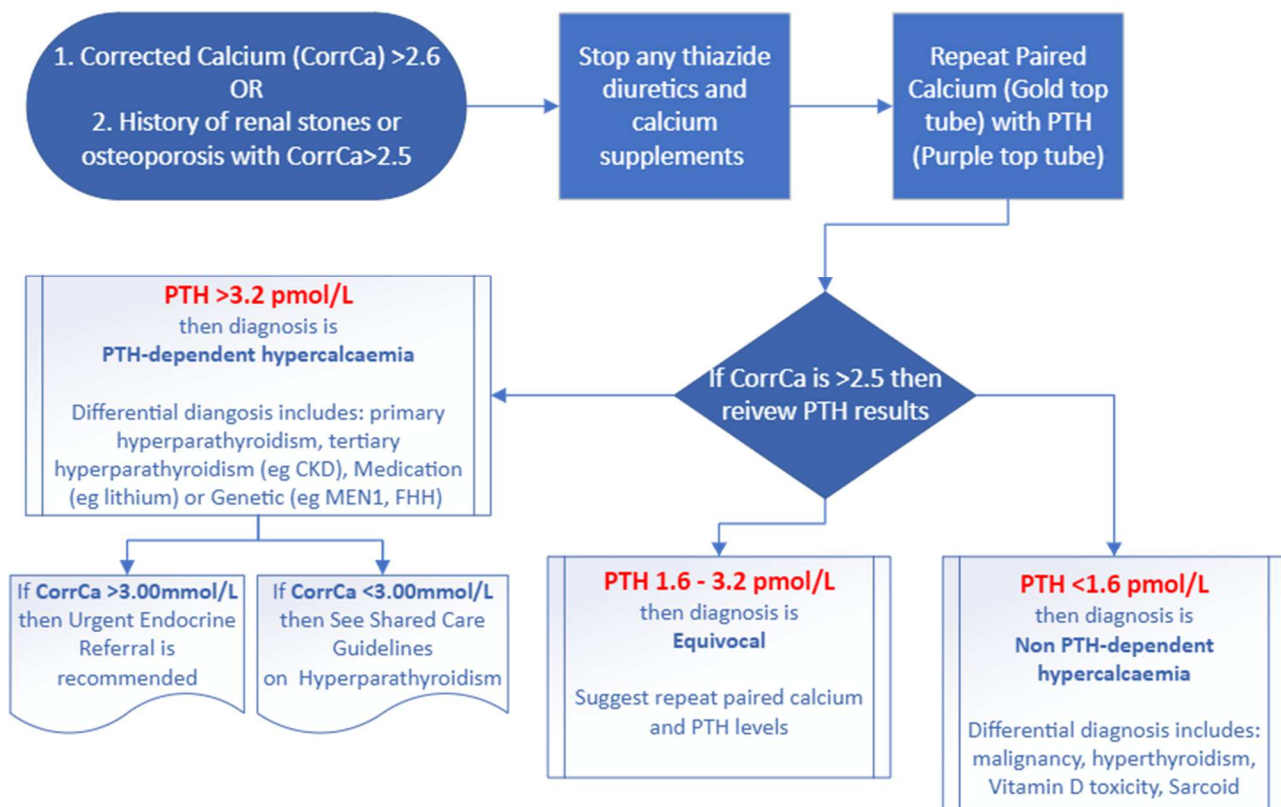
What are the major causes of hypercalcaemia?

- Primary Hyperparathyroidism
- Drug related including drugs (Lithium, Thiazide)
- Linked to Renal Disease
 - Tertiary Hyperparathyroidism
 - Treatment with Vitamin D analogues
- Malignancy
 - Primary Solid Tumour
 - Secondary in Bone
 - Haematological Malignancy

>90% of cases

Less common causes of hypercalcaemia

- Hyperthyroidism
- Sarcoid
- Vitamin D Toxicity



PTH-Dependent Hypercalcaemia

- **If calcium ≥ 3.00 mmol/L:** Urgent referral to Endocrinology is recommended
- **If calcium < 3.00 mmol/L:** See Shared Care Primary Hyperparathyroidism guideline for 'When to refer'
- For patients with lithium-dependent hypercalcaemia it is not recommended to stop the lithium treatment immediately, and each patient should be considered on a case by case basis. Recommend obtaining Endocrine advice by choosing 'Advice and Guidance' from NHS e-Referrals or see contacts at the end of this guideline.

Non PTH-Dependent Hypercalcaemia

Malignancy

This is the commonest cause of Non PTH-Dependent Hypercalcaemia and should be excluded as an urgent priority.

Three main types:

- Solid Tumours, PTHrP related, e.g. Lung cancer, Liver cancer, etc
- Haematological Malignancy, e.g. Myeloma
- Skeletal Metastases, e.g. Secondary to Breast cancer

For patients with known cancer suggest urgent referral to patient's hospital team. When the underlying malignancy is not known, referral will obviously depend upon underlying symptoms, history and other findings that are available at the time.

Hyperthyroidism

For hypercalcaemia associated with hyperthyroidism, suggest urgent referral to Endocrinology.

Vitamin D toxicity

Hypercalcaemia caused by "over the counter" Vitamin D preparations need to have a 25-Hydroxy Vitamin D level measured.

Most patients with hypercalcaemia caused by Vitamin D analogues (e.g. calcitriol) should be referred to the hospital team who first prescribed the medication.

Other causes

Unless the patient already has a definitive diagnosis (e.g. Sarcoid, or lymphoma) it is worthwhile sending "Advice & Guidance" to Endocrinology for patients with hypercalcaemia who do not fall into any of the above categories.

A final point

Primary Hyperparathyroidism is a relatively common disorder, therefore some patients may have more than one aetiology for the hypercalcaemia and/or have an underlying malignancy present.

Contacts

Derby

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131
Endocrinology Advice (urgent)	07879 115507 (9am – 5pm, Mon – Fri)
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Chesterfield

Duty Biochemist
On Call Consultant Biochemist
Endocrinology Advice (urgent)
Endocrinology advice (non-urgent)

01246 512212 (9am to 5pm, Mon – Fri)
Via CRH switchboard, 01246 277271
01246 513104
Choose 'Advice and Guidance' from
NHS e-Referrals

Burton

Duty Biochemist
Endocrinology advice

01283 566333 ext 4233
Via endocrine secretaries 01283 593259

References

1. Hyperparathyroidism (primary): diagnosis, assessment and initial management. NICE NG132 (May 2019)

Document title	Hypercalcaemia		
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Review history	Version/Date	Authors	Reason for review
	V1 August 2011	Dr N Lawson	New Document
	V2 Oct 2013	Dr R Rea, Dr P Blackwell, Mrs H Seddon	Routine review
	V3 March 2016	Dr R Stanworth, Dr P Blackwell, Mrs H Seddon	Routine review
	V4 May 2018	Dr R Stanworth, Dr P Blackwell, Mrs H Seddon	Routine review
	V5 Feb 2020	Dr P Blackwell, Dr R Stanworth, Dr R Robinson, Mrs H Seddon	Routine review
	V6 Feb 2023	Dr P Blackwell, Dr R Robinson, Dr D Hughes, Dr A Ugur, Mrs H Seddon	Routine review
	V7 May 2024	Dr P Blackwell, Dr R Robinson, Dr D Hughes, Dr A Ugur, Mrs H Seddon	Update of measuring units for PTH
	V8 May2025	Dr P Blackwell, Dr R Robinson, Dr P Shillo, Dr A Pillai Dr D Hughes, Dr A Ugur, Ms S Knowles	Routine review
Approval	Date	Name(s)	Designation
Secondary Care UHDB	23.04.25	Dr R Stanworth, Dr A Ugur, Dr H Ali, Dr D Hughes	Consultant Endocrinologists
Secondary Care CRH	01.05.25	Dr A Pillai, Dr R Robinson, Dr P Shillo	Consultant Endocrinologists
Primary Care	02.05.25	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	23.4.2025	Ms S Knowles	Consultant Clinical Scientist